Battery SOC and Remaining Charge-Time Estimation from Li-ion Drive-Cycle Data

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GitHub: https://github.com/SaiGanesh112006/TIH_VUDA_JAYA_SAI_GANESH/tree/main/

Problem and Objectives

- Estimate SOC (%) from voltage, current, temperature, power, ambient temperature
- Estimate remaining charging time (s) in CC-CV segments
- Leakage-safe evaluation: split by file, not by row
- Compare baselines vs Random Forest, XGBoost, MLP
- Deliver a frozen, reproducible package

Project Timeline

- Week 1: problem definition, literature review (8 methods), dataset design, pipeline plan
- Week 2: data audit (208 files), cleaning, preprocessing, frozen split, feature engineering
- Week 3: SOC baselines, ensembles, MLP, tuning; remaining charge-time target built (Day 17) and tuned
- Week 4: model selection by grouped CV, final test of SOC and charge time, failure analysis, package, report

Data Quality Audit

Check	Finding	Decision
Files / rows	208 CSVs, 6 temperatures, ~4.95M rows	Combine; later use 25°C folder
Duplicates	5,876 rows (0.12%)	Drop exact duplicates
Invalid voltage	3 files (non-positive)	Flag, do not auto-delete
Current outliers	6.75% (IQR)	Flag; high-load pulses
SOC column	Absent in raw data	Derive via coulomb counting

Preprocessing

- 25°C folder, 36 files; every 10th row: 104,560 x 17, 0 missing, 0 duplicates
- Target: SOC = 100 x (1 + capacity_Ah / 3.0), clipped 0-100
- Inputs: voltage_v, current_a, cell_temp_c, power_w, ambient_temp_c (no scaling)
- Leakage columns excluded: capacity_ah, wh_accu, cnt, step, cycle, status, time columns
- Charge-time target: time to end of each CHA segment

Train / Test Split

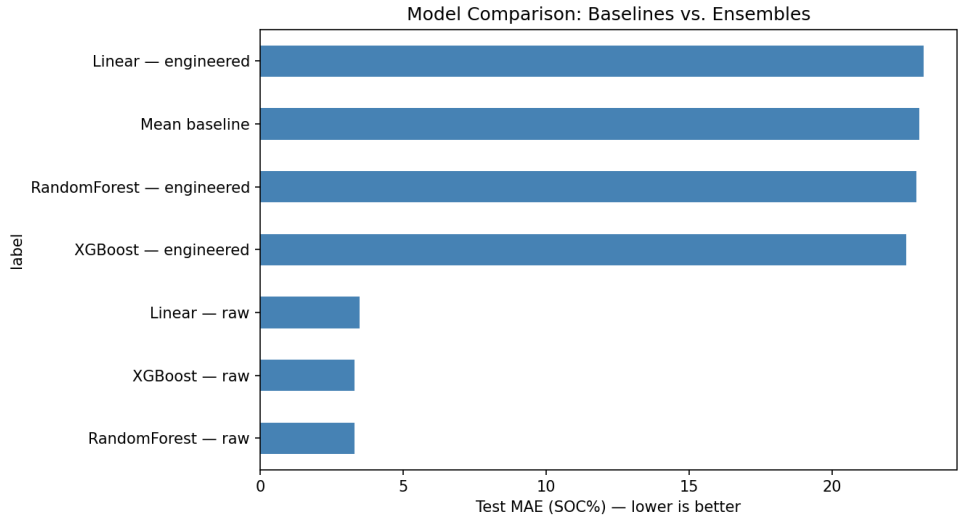
- GroupShuffleSplit by file_id, test_size 0.2, random_state 42 (frozen Dataset V1)
- Random row split rejected: correlated rows leak across train and test
- Week 2-3: 76,792 train / 27,768 test rows
- Week 4 frozen set: 5,356 rows; 18 train files (5,028 rows), 5 held-out files (328 rows)
- Tuning and selection on training files only; test scored once
- Charge-time target: same file groups; 9 train / 5 held-out files with charging

Feature Engineering

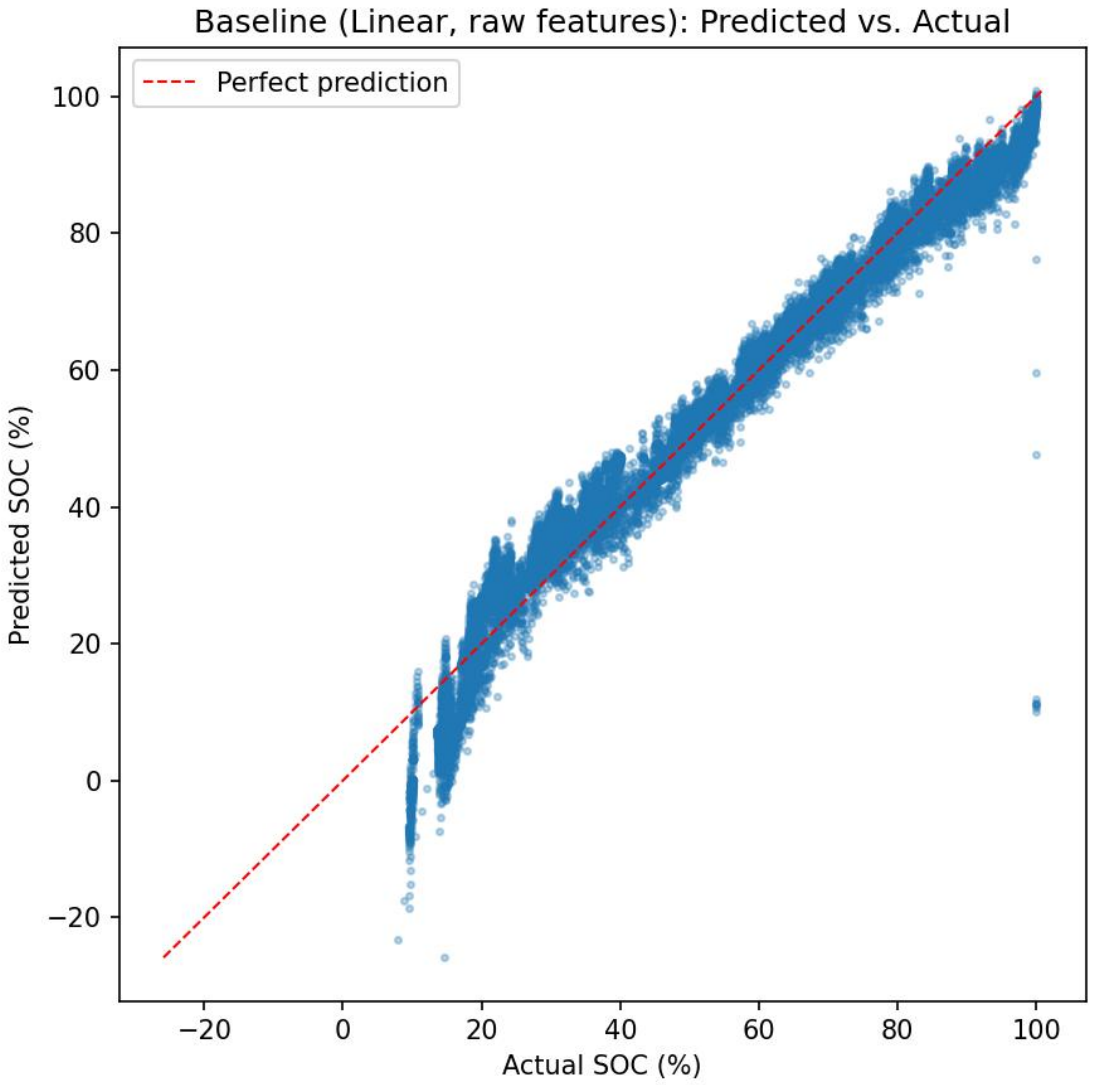
- Lags, deltas, rolling stats, C-rate, temp rise, interactions
- Pruned by correlation > 0.9 and VIF > 10 -> voltage_roll_std5, current_roll_std5
- Result: Linear MAE 3.47 -> 23.19; RF 3.16 -> 22.90
- Decision: engineered features rejected

Model Comparison (Week 3)

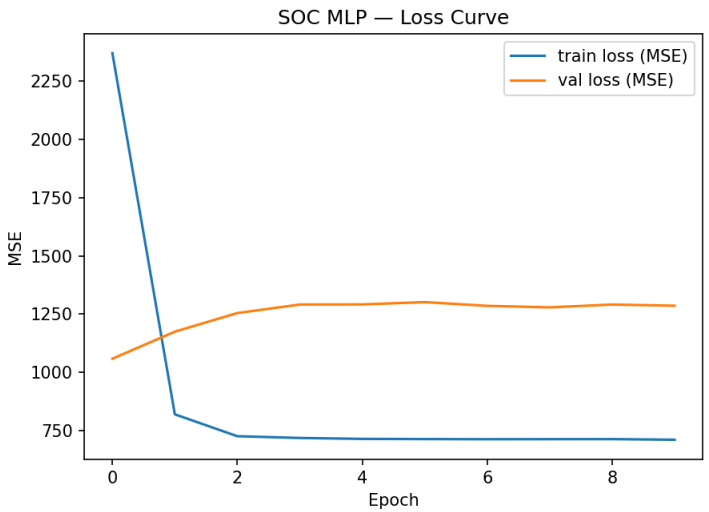
Method	Features	MAE (%)	RMSE (%)	R^2
Mean baseline	-	23.04	27.51	-0.048
Linear Regression	raw	3.47	4.59	0.971
Linear Regression	engineered	23.19	27.77	-0.068
Random Forest	raw	3.28	10.47	0.848
Random Forest	engineered	22.94	28.92	-0.158
Gradient Boosting (XGBoost)	raw	3.30	10.08	0.859
Gradient Boosting (XGBoost)	engineered	22.60	27.53	-0.049
MLP (ANN)	engineered	25.93	31.61	-0.384
RF (charging time)	charging	9,002 s	18,410 s	-0.160

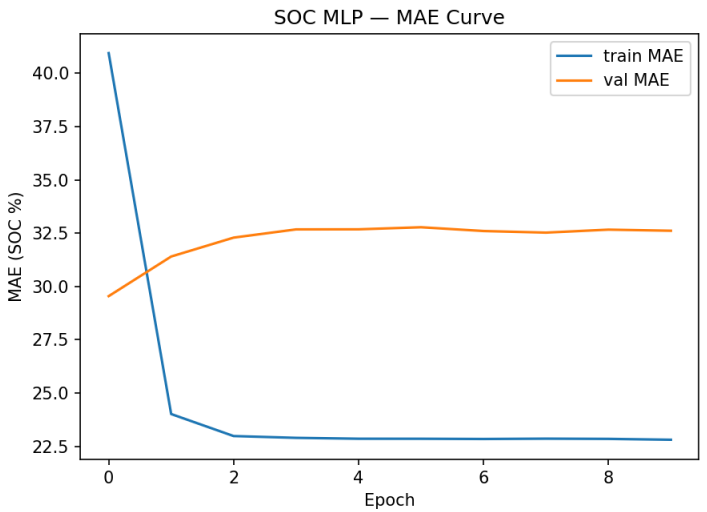


Predicted vs Actual (raw features)



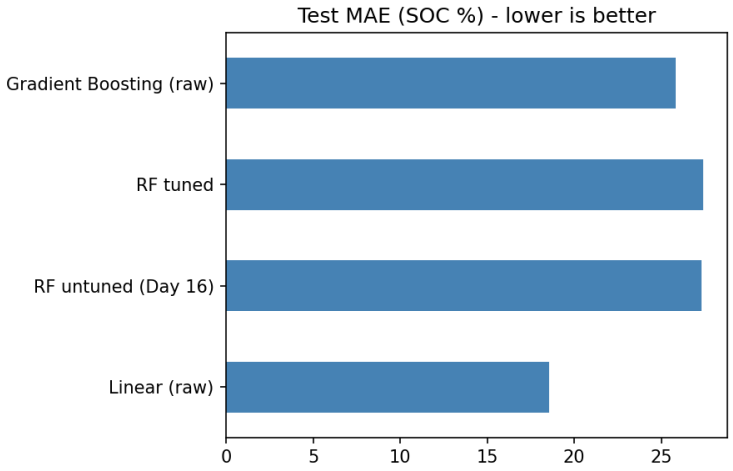
Neural Network (MLP)

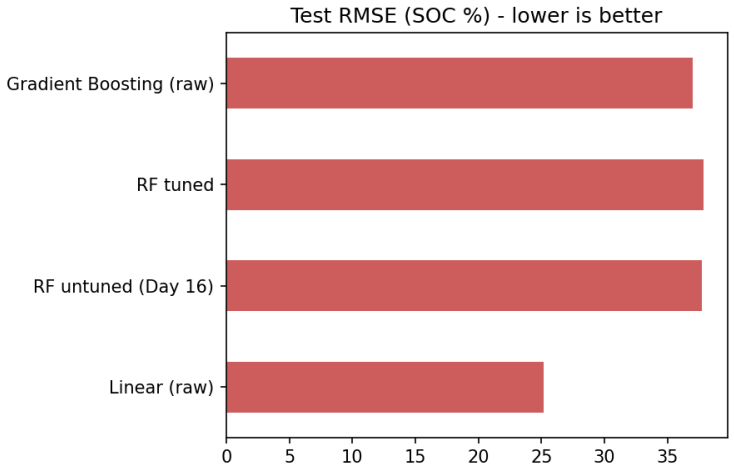




- 64-32-1, 2,305 parameters
- Early stopped at epoch 10
- MAE 25.93, R^2 -0.38 -> rejected

Hyper-parameter Tuning

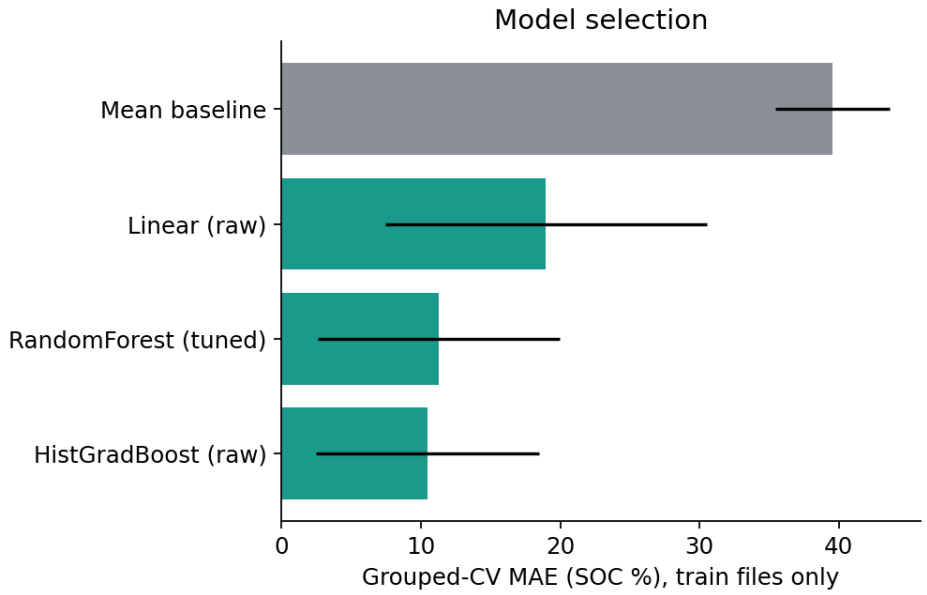




- Grid on grouped validation split
- Best RF: depth 8, leaf 4, 300 trees
- Early split: MAE 3.28 -> 2.90
- Frozen split: only -0.3% change

Model Selection (Grouped 5-fold CV)

Model	Rank	CV MAE	CV RMSE	CV R^2	Worst-fold MAE
Mean baseline	0	39.551	41.769	-8.591	44.118
Linear (raw)	1	18.972	29.117	-8.728	29.847
RandomForest (tuned)	2	11.292	20.415	-1.312	26.332
HistGradBoost (raw)	3	10.462	19.363	-1.369	24.125



- Rule fixed before test: within 10% of best CV MAE -> simplest
- Chosen: RandomForest (tuned)
- Flag: worst fold > 2x mean

Final Test on Held-out Files

Model	MAE (%)	RMSE (%)	R^2
Mean baseline	28.93	32.19	-0.033
Linear regression (raw)	18.10	25.16	0.369
RandomForest (tuned) - frozen final	27.73	37.71	-0.417

- RF is worse than Linear on unseen files (R^2 -0.417)
- Linear: 1.3 ms, 0.001 MB vs RF 89.6 ms, 6.1 MB
- Recommendation: revert to the simple baseline
- Charge time: RF MAE 26,626 s vs mean 27,124 s (see next slides)



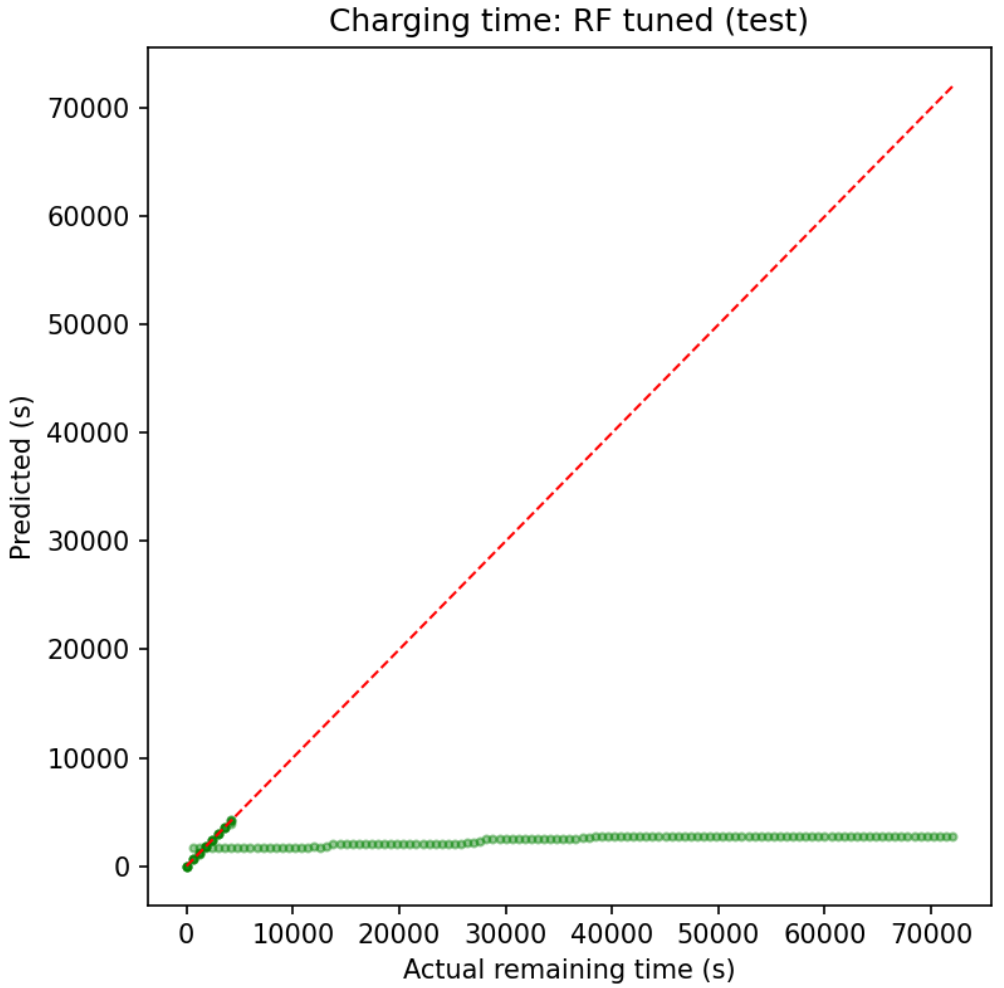
Remaining Charge Time: Method

- Target: seconds from each row to the end of its contiguous CHA (charging) segment
- Data: 1,299 charging rows, 67 segments; inputs voltage, current, cell temp, ambient temp
- ANN not used: too few, repetitive CC-CV segments (would memorise)
- Random Forest chosen; tuned to max_depth 10, 100 trees (Day 19)
- Early split: MAE 9,002 s; frozen held-out: 153 rows from 5 files

Remaining Charge Time: Results

Model	MAE (s)	MAE (min)	RMSE (s)	R^2
Mean baseline	27,124	452.1	35,448	-1.336
Linear	27,021	450.3	35,219	-1.306
Random Forest (tuned)	26,626	443.8	35,060	-1.285

- RF MAE 26,626 s (~444 min), only slightly better than mean baseline
- Low confidence: indicative only

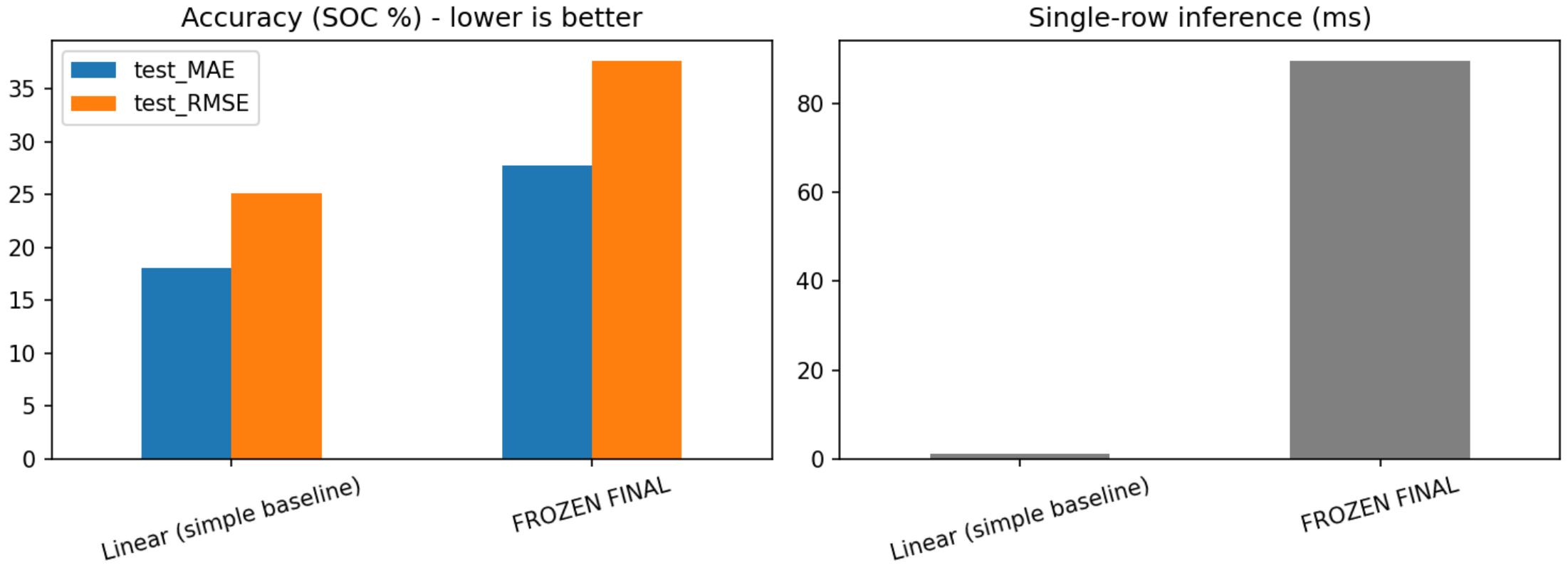


Remaining Charge Time: Error by File

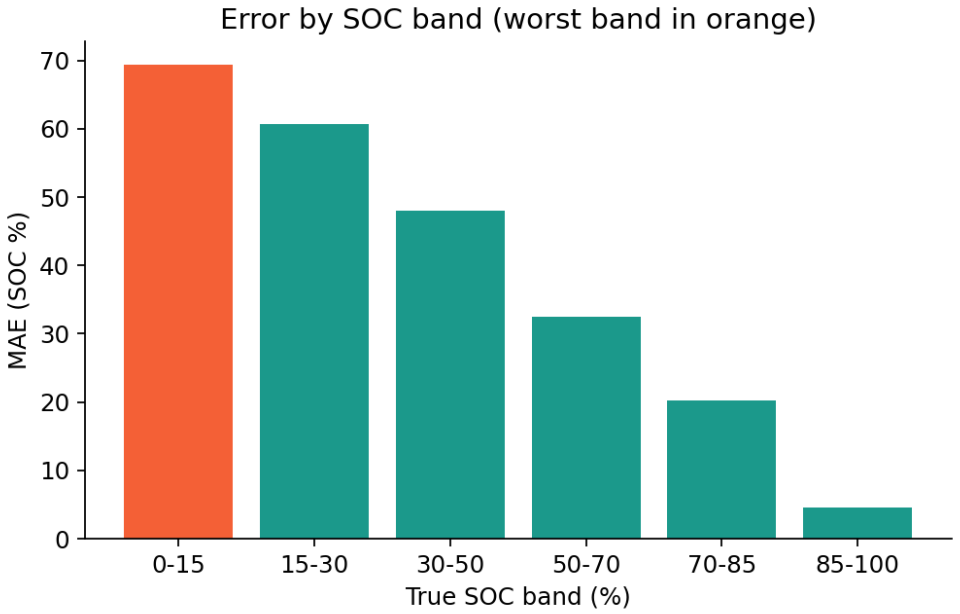
Held-out file	Rows	MAE (s)	Max error (s)
25degC_C20DisCh	121	33,663	69,276
25degC_Charge14	8	39.0	270
25degC_Charge15	8	13.9	42
25degC_Charge6	8	1.9	12
25degC_Charge8	8	6.8	42

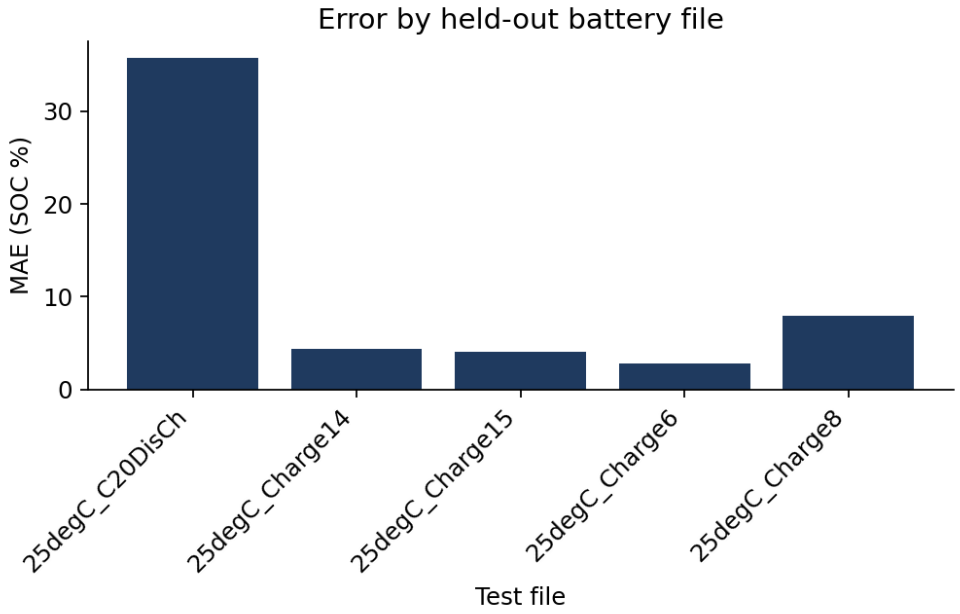
- 4 true charge files: MAE only 2 to 39 s
- C20DisCh (121 rows tagged CHA) has MAE 33,663 s and dominates the aggregate
- Needs more charge segments and a consistent segment definition

Accuracy vs Complexity



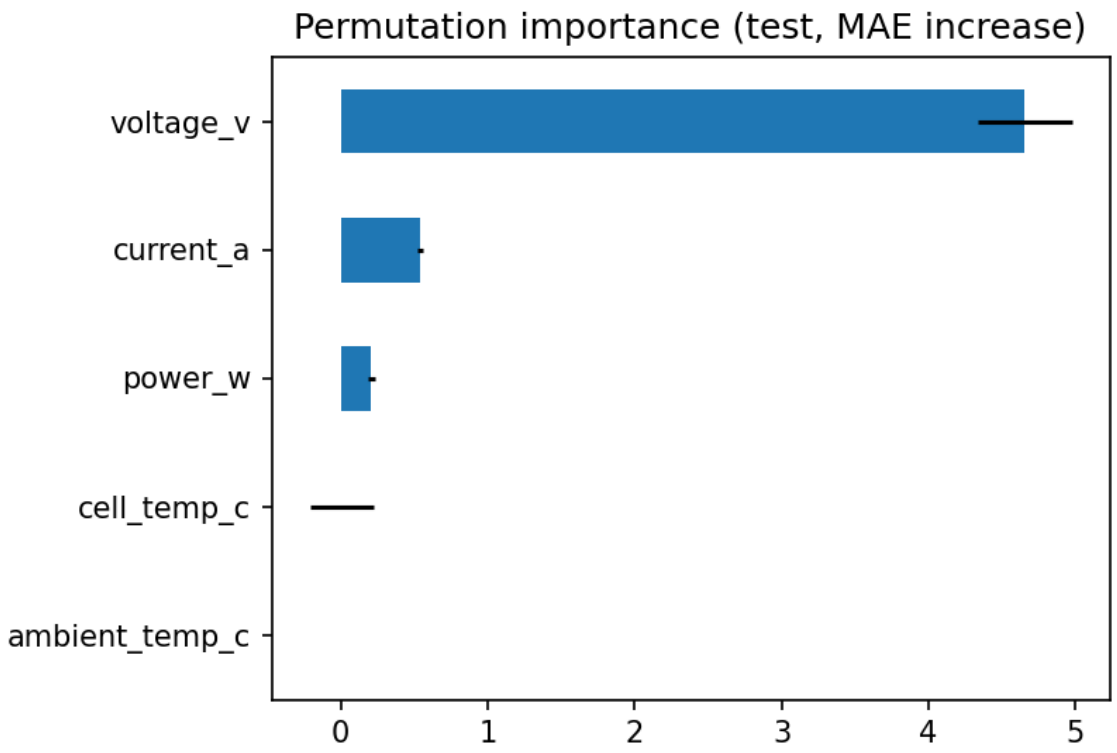
Error Analysis: SOC Bands and Files

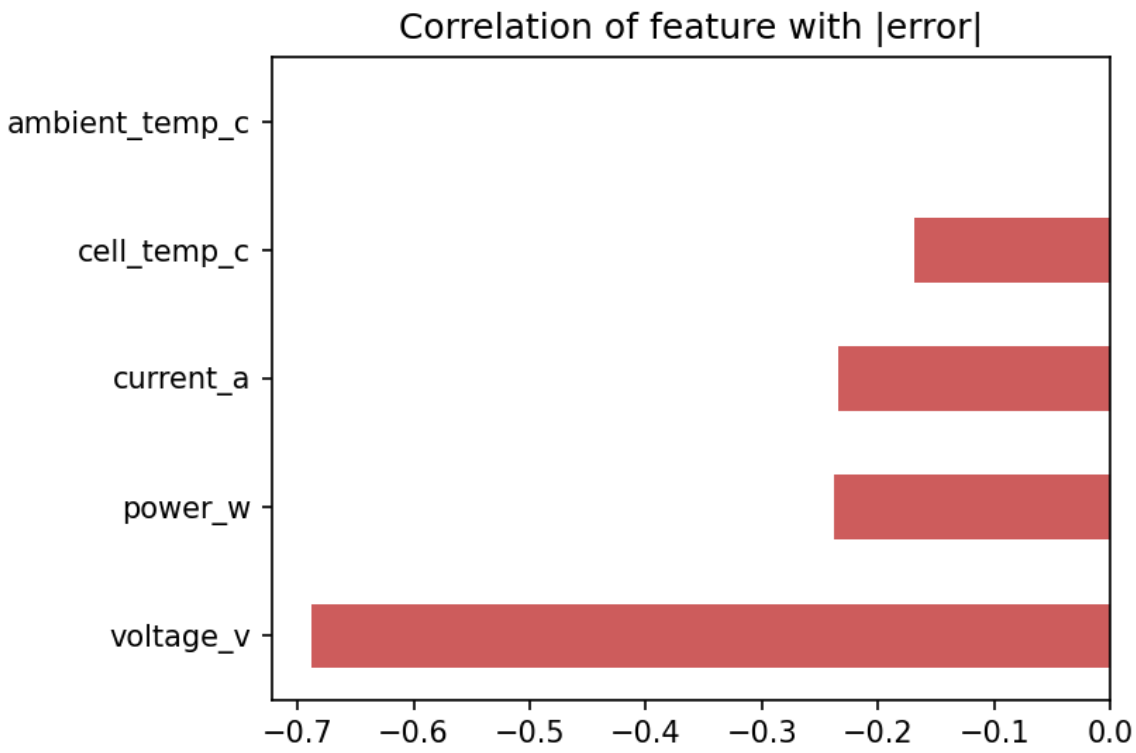




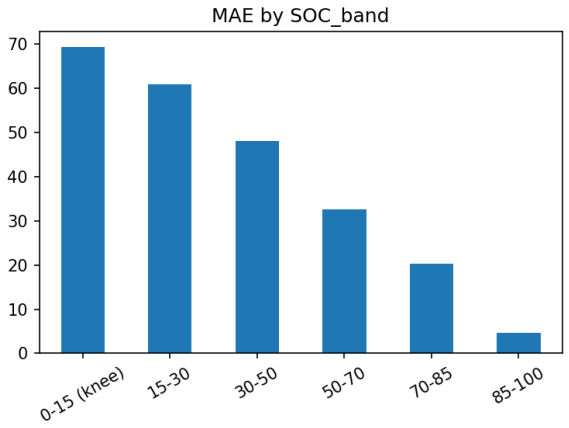
Held-out file	MAE (%)	Max error	Rows
25degC_C20DisCh	35.74	92.69	243
25degC_Charge8	7.90	85.95	22
25degC_Charge14	4.33	32.92	21
25degC_Charge15	4.10	47.32	21
25degC_Charge6	2.77	22.45	21

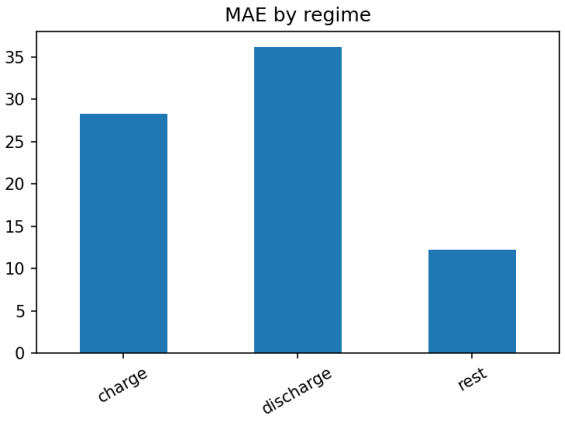
Feature Importance

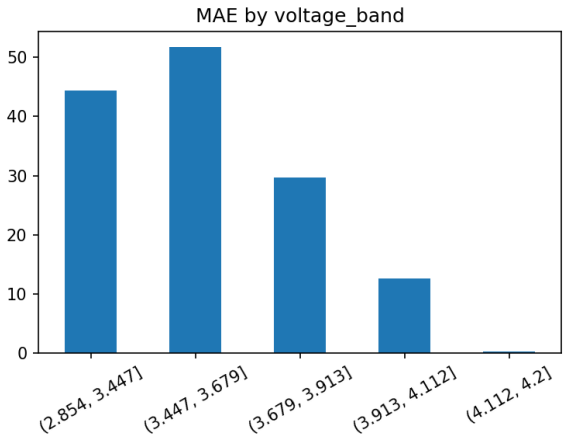


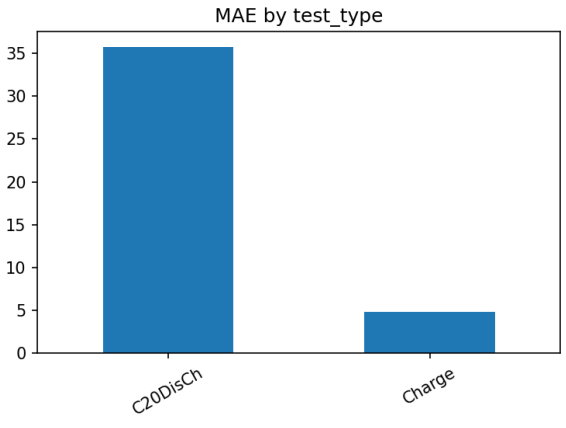


Errors by Operating Condition









Limitations

- Only 5 held-out files / 328 rows
- Worst at SOC 0-15% (MAE 69%); one discharge file carries ~97% of squared error
- Charge time: only 5 files, one long file dominates error
- Only 25°C data; no history in model; V, I, P collinear
- Early MAE ~3% results were on a larger, easier split; not comparable with final

Software Package and Reproducibility

- battery_soc module: config, data, models, train, evaluate, predict
- Frozen models + frozen_config.json, pytest tests, example I/O
- predict.py --model-dir models --input data/sample_input.csv
- Repo: https://github.com/SaiGanesh112006/TIH_VUDA_JAYA_SAI_GANESH/tree/main/

Conclusion and Future Work

- Leakage-safe grouped evaluation exposed that optimistic MAE ~3% does not generalise
- SOC: Linear on raw features (MAE 18.1%) beats tuned RF (27.7%) on unseen files
- Remaining charge time: RF MAE 26,626 s overall; 2 to 39 s on standard charge files; needs more segments
- Next: GRU/LSTM sequence models, multi-temperature data, uncertainty estimates, more charge segments